

Failure Modes & Honest Degradation

AirOne is built around one principle: **never present fabricated, stale, or fake-complete data as real**. This document defines the vocabulary the system uses to tell the truth about its own state, and how each subsystem degrades.

Quality vocabulary (QualityState)

Every Measurement carries a QualityState:

State	Meaning
VALID	Trustworthy, in-range, current.
SUSPECT	Passed basic checks but a rate/consistency check flagged it — treat with caution.
INVALID	Failed a hard check (range, schema). The numeric value must not be used.
MISSING	The field was expected but absent.
STALE	Older than the freshness bound; no longer current.
CORRUPTED	Failed CRC or FEC could not recover it.
REPAIRED	Recovered via FEC; usable but flagged as reconstructed.
SIMULATED	Produced by the simulator, not a real sensor.
ESTIMATED	Derived/inferred rather than directly measured.

An INVALID/MISSING/CORRUPTED measurement is forced to valid = False by the model itself — the flag and the state can never disagree.

Provenance vocabulary (DataSource)

RAW, CALIBRATED, FILTERED, FUSED, ESTIMATED, SIMULATED. Every value knows how it was produced, so a fused or estimated number is never mistaken for a raw sensor reading.

Tier-level degradation

Tier	Failure	Observable result
1 Raw	Serial port missing	Receiver reports the port error; the system does not invent telemetry. Use --simulate for demos.

Tier	Failure	Observable result
1 Raw	CRC failure	Frame marked CORRUPTED; not analysed.
2 Science	Too few samples	Analysis returns an explicit insufficient/unavailable result, not a guess.
2 Science	One analysis throws	That analysis is marked failed; the other 36 and all of Tier 1 continue.
3 ML	sklearn/torch absent	Those detectors report available=false with the import error; robust_zscore still works.
3 ML	Worker not running	/ml/anomalies returns 503 NOT_CONFIGURED.
3 ML	Model file tampered	ModelStore.load raises SafeLoadError; the model is not loaded.
4 GUI	No display / PyQt5 absent	Launcher reports it and runs headless; backend unaffected.
4 GUI	API unreachable	Views show the error and the tier state; no cached/fake numbers are shown as live.

API-level honesty

- Endpoints that depend on a live worker return 503 with data.state = "NOT_CONFIGURED" when that worker is absent.
- Requests that are well-formed but cannot be honestly fulfilled (e.g. training with no data) return 422 with a precise reason — never a fabricated success.
- The response envelope always includes success, error, and often a state, so a client can always distinguish “no data” from “data is zero”.

Secrets & startup

If AIRONE_JWT_SECRET is unset or is a known placeholder, the system refuses to start (InsecureConfigError). This is a fail-closed control: it is better to not start than to run with a guessable secret.

The anti-pattern this system avoids

The following are explicitly **never** done anywhere in the codebase:

- Returning 0.0 (or any placeholder) for a failed sensor instead of INVALID.
- Imputing missing features with fabricated values before ML/analysis.

- Showing the last-known value as if it were live when it is actually STALE.
- Reporting a PSI of 0 when there is not enough data to compute one.
- Claiming a detector/model works when its dependency is missing.